1. Supervised Learning

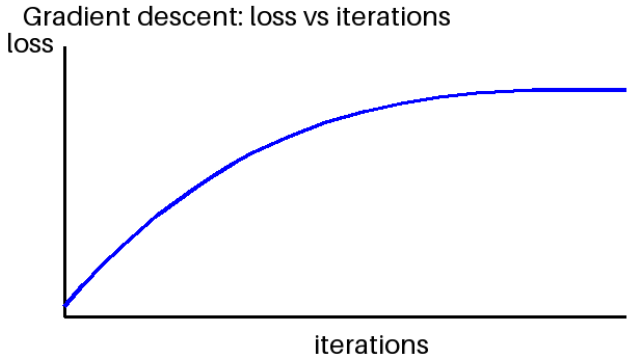
Supervised learning trains a model on labelled examples, where each input x is paired with a target output y. The goal is to learn a function f that predicts y for unseen inputs. Regression predicts a continuous value such as house price, while classification predicts a discrete label such as spam or not spam. The labelled dataset is split into a training set, used to fit the model, and a test set, used to measure generalisation.

2. Linear Regression

Linear regression models the target as a weighted sum of the inputs plus a bias term, y = w1x1 + w2x2 + b. The weights are chosen to minimise the mean squared error, the average of the squared differences between predictions and true values. For one feature the fitted model is a straight line through the data. The coefficient of determination R squared measures the fraction of variance explained by the model.

3. Gradient Descent

Gradient descent is an iterative optimisation algorithm that minimises a loss function. At each step the parameters move in the direction opposite to the gradient, scaled by the learning rate alpha: w = w - alpha times the gradient. If the learning rate is too large the updates overshoot and the loss diverges, whereas a very small learning rate converges slowly. Stochastic gradient descent estimates the gradient from a small mini-batch of 32 examples instead of the full dataset.

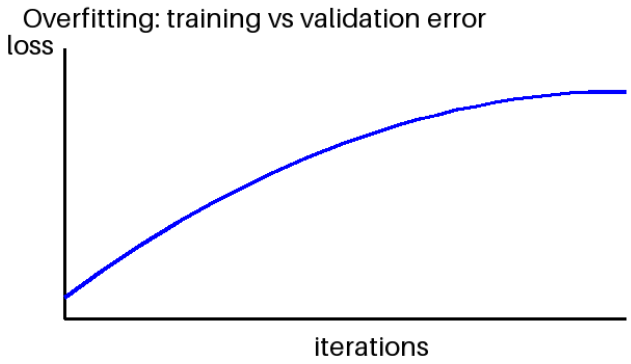


4. Logistic Regression and Classification

Logistic regression is a classification model that passes a linear score through the sigmoid function, sigma(z) = 1 / (1 + e^-z), to produce a probability between 0 and 1. A decision threshold, commonly 0.5, converts the probability into a class label. The model is trained by minimising the cross-entropy loss, also called log loss. The decision boundary is the set of points where the predicted probability equals the threshold.

5. Overfitting and Regularization

Overfitting occurs when a model fits the noise in the training data and therefore performs poorly on new data. It is detected when training error is low but validation error is high. Regularization discourages complexity by adding a penalty to the loss: L2 regularization, also called ridge, penalises the sum of squared weights, while L1 regularization, called lasso, penalises the sum of absolute weights and can drive some weights exactly to zero. Cross-validation with 5 folds is a standard way to choose the regularization strength.



6. Bias-Variance Tradeoff

The expected prediction error of a model decomposes into bias squared, variance, and irreducible noise. High bias means the model is too simple and underfits, missing real structure in the data. High variance means the model is too flexible and its predictions change greatly between different training sets, which is overfitting. Increasing model complexity lowers bias but raises variance, so the best generalisation lies at an intermediate complexity.

7. Neural Networks and Backpropagation

A neural network stacks layers of neurons, where each neuron computes a weighted sum of its inputs and applies a non-linear activation function such as ReLU, defined as max(0, z). Without non-linear activations a deep network would collapse into a single linear model. Backpropagation computes the gradient of the loss with respect to every weight by applying the chain rule backwards from the output layer to the input layer. These gradients are then used by gradient descent to update the weights.

8. Evaluation Metrics

Accuracy is the fraction of correct predictions but is misleading on imbalanced data. Precision is the fraction of predicted positives that are truly positive, while recall is the fraction of actual positives that are found. The F1 score is the harmonic mean of precision and recall. For example, a classifier with precision 0.8 and recall 0.5 has an F1 score of about 0.62. The ROC curve plots the true positive rate against the false positive rate across thresholds.

9. Clustering with K-Means

K-means is an unsupervised algorithm that partitions data into k clusters. It starts by choosing k initial centroids, then alternates between assigning every point to its nearest centroid and recomputing each centroid as the mean of its assigned points. The algorithm stops when assignments no longer change. The result depends on initialisation and on the choice of k, which can be guided by the elbow method that plots within-cluster variance against k.